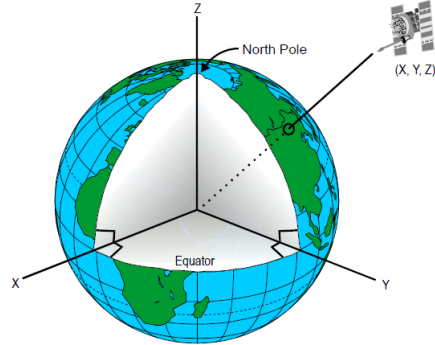


Orbital Geometry Guide

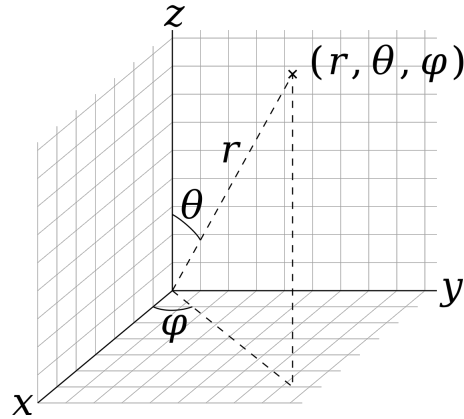
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Abstract



(a) Earth-center, Earth-fixed reference frame (ECEF).



(b) Polar (θ, ϕ) reference frame.

Figure 1: Comparison of the ECEF and Polar reference frames.

Coordinate Systems

ECEF, Polar, & Latitude-Longitude Coordinates

In Figure 1 (a) of the **ECEF** reference frame, the z^+ -axis is the North Pole, x^+ passes through the intersection of the Equator (0° latitude) and Prime Meridian (0° longitude), and the y^+ -axis passes through the Equator and 90° longitude arc.

Figure 1 (b) shows us that in the **polar** coordinate system, θ is the angle measured down from the z^+ -axis toward the xy -plane, and ϕ is the angle measured from the x^+ -axis, counterclockwise.

The only difference between the polar coordinate system and the **latitude-**

longitude coordinate system is that a point's latitude is measured up from the xy -plane towards the z^+ axis. In order to find the **altitude** we simply take

$$\text{Alt} = r - r_e$$

, where r_e is the radius of the Earth at that point. (Note: the Earth is elliptical, but for the sake of the calculations in this guide, we will be assuming a spherical Earth with radius $r_e = 6378$ km.)

The conversions between these frames is as follows:

Polar-to-LLA

$$\text{Latitude} = \Theta = 90 - \theta$$

$$\text{Longitude} = \Phi = \phi$$

ECEF-to-LLA (and Polar!)

$$r = \sqrt{x^2 + y^2 + z^2}$$

$$\text{Alt} = r - r_e$$

$$\theta = \cos^{-1}\left(\frac{z}{r}\right)$$

$$\text{Latitude} = 90^\circ - \cos^{-1}\left(\frac{z}{r}\right)$$

$$\text{Longitude} = \phi = \tan^{-1}\left(\frac{y}{x}\right)$$

LLA-to-ECEF

$$x = r \cos(\text{Latitude}) \cos(\text{Longitude})$$

$$y = r \cos(\text{Latitude}) \sin(\text{Longitude})$$

$$z = r \sin(\text{Latitude})$$

Polar-to-ECEF

$$x = r \sin(\theta) \cos(\phi)$$

$$y = r \sin(\theta) \sin(\phi)$$

$$z = r \cos(\theta)$$

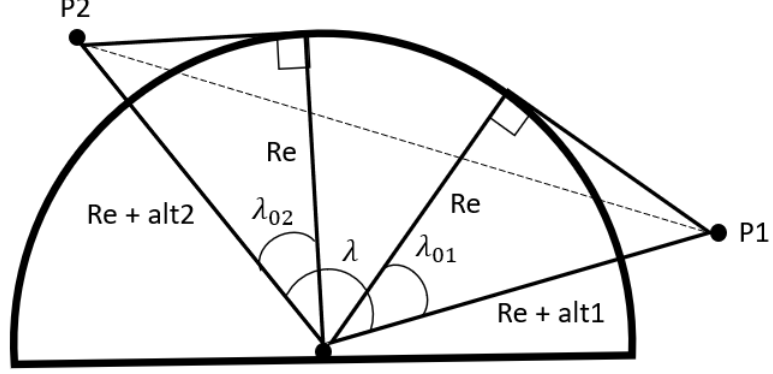


Figure 2: Illustration of the relationship between λ_{01} , λ_{02} and λ

Platform-centered Coordinate Systems (PCCS)

The local coordinate system of the platforms has the **nadir** vector (that is, the vector pointing from the center of the platform to the center of Earth) as the z^+ -axis. The sub-platform point (SPP) is the point at which the nadir line intersects the surface of the Earth. In the PCCSs, the x^+ -axis parallel to, and directionally the same, as the northward vector of the SPP. For our purposes, the directionality of the y -axis is ambiguous.

Line of Sight (LOS)

First, in order to determine if two platforms have LOS, we need to calculate the λ_0 (the angular region of Earth) seen by each platform. Clearly, we can see that λ_0 for each platform is given by

$$\lambda_{01} = \frac{r_e}{r_e + alt_1}$$

$$\lambda_{02} = \frac{r_e}{r_e + alt_2}$$

Note that for a platform on the Earth's surface, $\lambda_0 = 0$.

Then we need to find λ , which is the inner-Earth angle between the platforms. We can accomplish this by leveraging the Spherical Law of Cosines, which states

$$\cos c = \cos a \cos b + \sin a \sin b \cos C$$

in a situation analogous to Figure 3 [3].

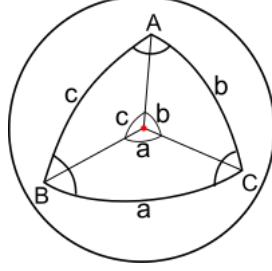


Figure 3: The Spherical Law of Cosines.

In the case of platforms on or orbiting Earth, we can leverage this by looking at the latitude and longitude of the SPPs. These projections will have the same angular relationships as the platforms themselves. So, The Spherical Law of Cosines becomes

$$\cos \lambda = \cos \theta_1 \cos \theta_2 + \sin \theta_1 \sin \theta_2 \cos(|\phi_1 - \phi_2|)$$

or in terms of latitude and longitude:

$$\cos \lambda = \sin \text{lat}_1 \sin \text{lat}_2 + \cos \text{lat}_1 \cos \text{lat}_2 \cos \Delta L$$

where $\Delta L = |\text{lon}_1 - \text{lon}_2|$.

Now, we can compare the λ_0 s to λ . If $\lambda \leq \lambda_{01} + \lambda_{02}$, then there is LOS. Otherwise, there is no LOS between the platforms.

As an alternative, we can use the Definition of the Inner Product and the ECEF coordinates found for each platform in order to calculate λ . We can think of the ECEF coordinates as the directional vector for each platform, originating at the center of the Earth. Therefore, using the Dot Product, we observe

$$\vec{p}_1 = \begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix}$$

$$\vec{p}_2 = \begin{pmatrix} x_2 \\ y_2 \\ z_2 \end{pmatrix}$$

$$\mathbf{p}_1 \cdot \mathbf{p}_2 = \sum_{i=1}^n p_{1i} p_{2i} = \|\mathbf{p}_1\| \|\mathbf{p}_2\| \cos \lambda$$

which we can rearrange to

$$\cos \lambda = \frac{x_1 x_2 + y_1 y_2 + z_1 z_2}{\sqrt{x_1^2 + y_1^2 + z_1^2} \sqrt{x_2^2 + y_2^2 + z_2^2}}$$

Another factor in LOS is elevation, ϵ , of the line-of-sight above Earth's surface (not to be confused with the local **azimuth and elevation** "look" angles,

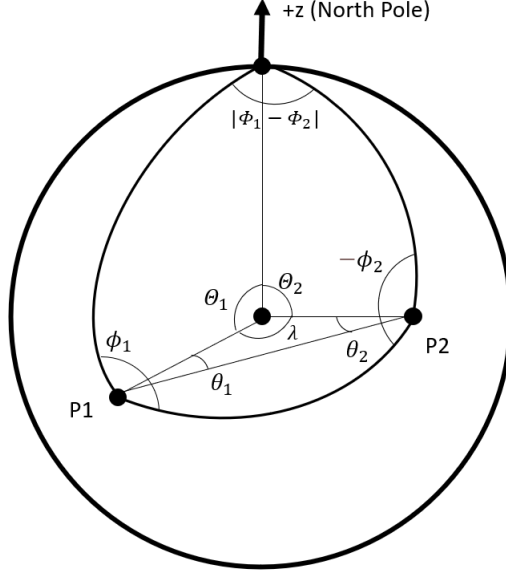


Figure 4: Illustration of the relationship of orbital angles, as they relate to the Spherical Law of Cosines.

which we will discuss later!) Many communication systems have a minimum elevation angle at which they can reliably operate. With respect to (w.r.t) platform (1), we see that ρ and λ_0 are related by

$$\cos \lambda_0 = \sin \rho = \frac{r_e}{r_e + alt_1}$$

(Note: if we were calculating w.r.t platform (2), we would use alt_2 instead of alt_1 in this calculation.)

We can calculate η as follows: We can see from the configuration in Figure 5 that

$$\sin \lambda = \frac{a}{r_e + alt_2} \implies a = (r_e + alt_2) \sin \lambda$$

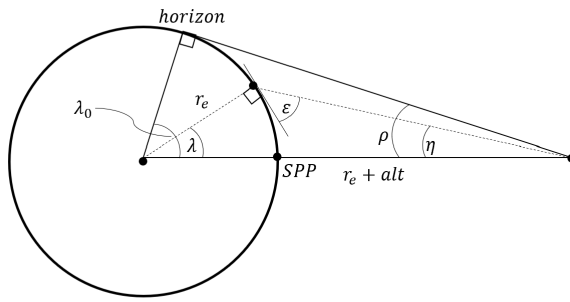
$$\cos \lambda = \frac{b}{r_e + alt_2} \implies b = (r_e + alt_2) \cos \lambda$$

$$(r_e + r_1) - b = (r_e + r_1) - (r_e + alt_2) \cos \lambda$$

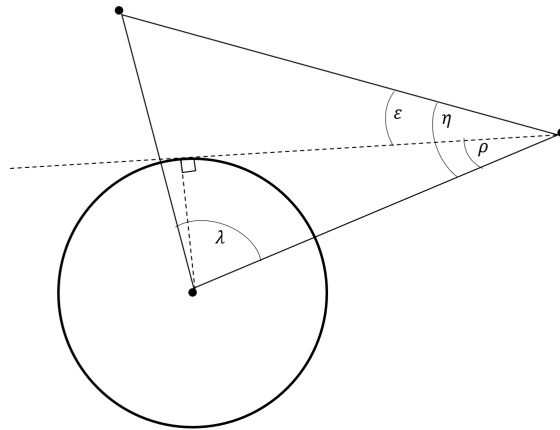
we can now use these relationships to determine

$$\tan \eta = \frac{(r_e + alt_2) \sin \lambda}{(r_e + r_1) - (r_e + alt_2) \cos \lambda}$$

Note, that is platform (2) is on the surface of the Earth, $r_e + alt_2 = r_e$, as



(a) Platform on surface of Earth.



(b) Platform above surface of Earth.

Figure 5: Geometry of ρ , η , and ϵ (i.e., elevation above local horizon.)

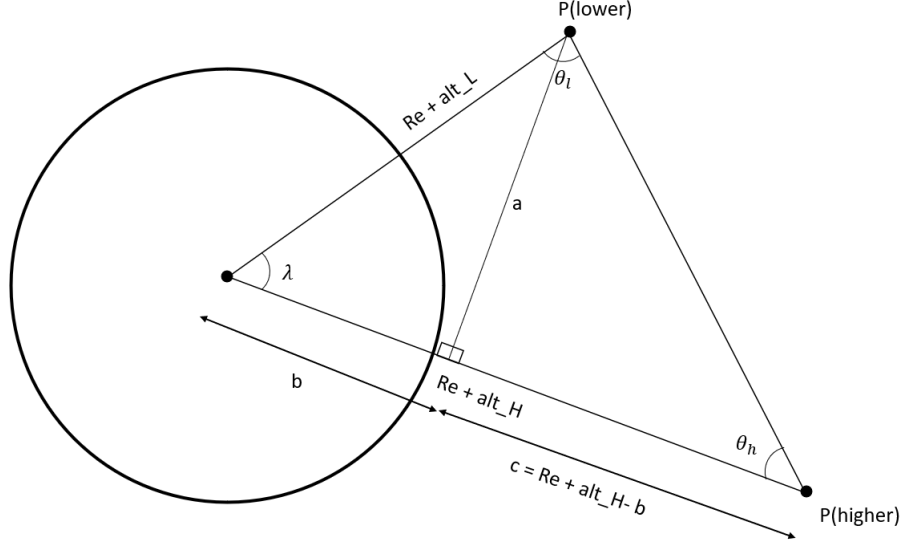


Figure 6: Illustration of the relationship of orbital angles, as they relate to the Spherical Law of Cosines.

$alt_2 = 0$, therefore

$$\tan \eta = \frac{r_e \sin \lambda}{(r_e + r_1) - r_e \cos \lambda} \left(\frac{1/r_e + r_1}{1/r_e + r_1} \right)$$

$$\tan \eta = \frac{\sin \rho \sin \lambda}{1 - \sin \rho \cos \lambda}$$

Moreover, we can also calculate η using the Law of Sines. Observe that

$$\frac{\sin \eta_1}{re + alt_2} = \frac{\sin \eta_2}{re + alt_1} = \frac{\sin \lambda}{D}$$

where D is the range between the platforms. Rearranging, we get the following relationships

$$\sin \eta_1 = \frac{(re + alt_2) \sin \lambda}{D}$$

$$\sin \eta_2 = \frac{(re + alt_1) \sin \lambda}{D}$$

$$\eta_2 = 180^\circ - \lambda - \eta_1$$

We can calculate D using the ECEF coordinates of each platforms:

$$D = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2}$$

From the Figures 5(a) and 5(b), we can tell that we have 2 cases.

(1) if $\rho > \epsilon$ (i.e., for a platform on Earth's surface),

$$\epsilon = 90^\circ - (\eta + \lambda)$$

or,

(2) if $\eta \geq \rho$ (i.e., for 2 airborne platforms),

$$\epsilon = \rho - \eta$$

More on LOS

Observe that we can derive the following relationships between η , λ , and ϵ from the above diagram

$$\cos \epsilon = \frac{\sin \eta}{\sin \rho} \implies \sin \eta = \frac{\cos \epsilon}{\sin \rho}$$

and the range between the two platforms, D , can be calculated from

$$D = r_e \frac{\sin \lambda}{\sin \eta}$$

or, more readily, using the ECEF coordinate frame

$$D = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2}$$

Now, clearly we will have the range early in our calculations. So, if we are given the minimum above Earth elevation angle, ϵ_{min} , we can calculate the maximum distance between platforms where they will still have LOS to one another, D_{max} , unobstructed by Earth. Observe:

$$\sin \eta_{max} = \sin \rho \cos \epsilon_{min}$$

$$\lambda_{max} = 90^\circ - \epsilon_{min} - \eta_{max}$$

$$D_{max} = r_e \left(\frac{\sin \lambda_{max}}{\sin \eta_{max}} \right)$$

And clearly, if $D \leq D_{max}$, the platforms have LOS.

This may reduce the number of calculations needed to find the necessary angles, given your needs!

Local Viewing Angles

This section discusses the viewing angles of one platform to another, with respect to its local **East-North-Up (ENU)** coordinate system.

East-North-Up (ENU).

The ENU coordinate system is simply the formal definition of a viewer's local horizon. For example, consider a viewer on the surface of Earth looking in the direction of an airborne platform. The East-North plane for the viewer is just the plane tangent to the surface of Earth, with one axis pointing true North, and the other, perpendicular axis pointing East (Note: to the Earth-bound viewer, these are just East and North, for an airborne viewer this is still true, but North and East are defined as the plane parallel to the East-North plane of the platform's SPP, containing the platform.) "Up" is the axis orthogonal to the North-East plane, which in the case just happens to be **zenith** (recall the zenith (the opposite of nadir) is the vector coming from the Earth's center, toward the platform).

Azimuth & Elevation.

Now that we have an understanding of the local ENU coordinate system for a platform, we can discuss the definitions of the viewing angles in the direction of another platform. The **azimuth** is the rotational angle, within the North-East plane; that is the angle off of North, in the direction of East. The **elevation** angle (sometimes called the "altitude angle") is then the angle measured "up" off the North-East plane (in the direction of the Up-axis, to be clear). From the diagram we observe the following (this has more meaning for Earth-bound platforms, but it is true regardless):

Cardinal Direction	Azimuth
North	0°
South	90°
East	180°
West	270°

Az-El Calculation Method #1: From Local Polar Coordinate Frame.

Calculating Range.

(1) Given that there is LOS to the airborne platform, we can first calculate the distance, D , between the two platforms, called the **range**. This is trivial, given we have the ECEF coordinates of the two platforms. Let

$$\Delta x = x_1 - x_2$$

$$\Delta y = y_1 - y_2$$

$$\Delta z = z_1 - z_2$$

$$D = \sqrt{\Delta x^2 + \Delta y^2 + \Delta z^2}$$

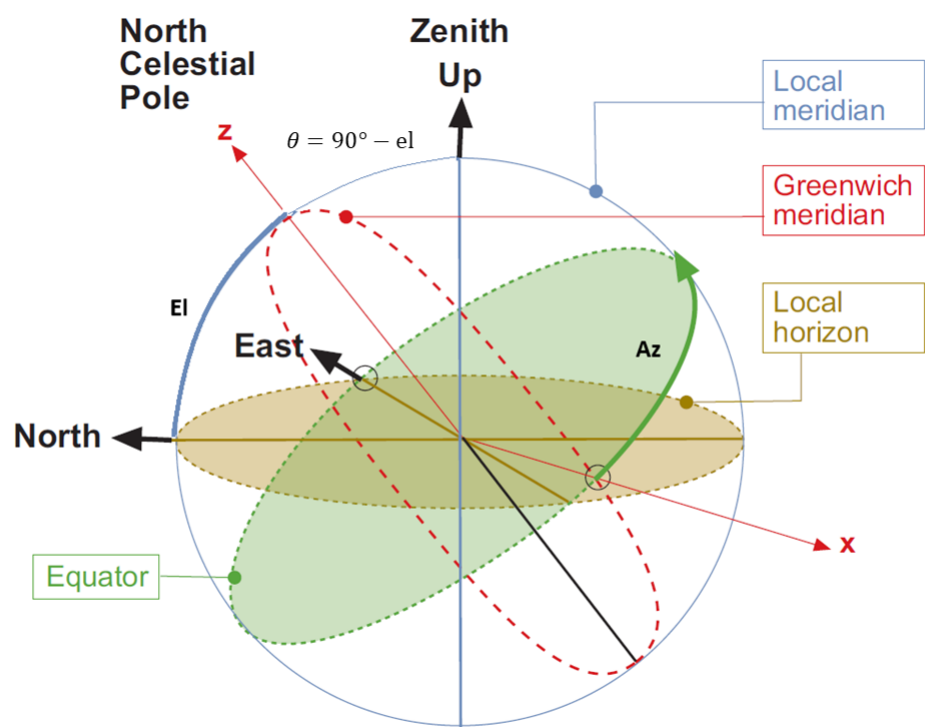


Figure 7: Azimuth & Elevation coordinate frame [1].

(2) Alternatively, we can calculate the range using angles we have already found using the following computation:

$$D = \frac{(r_e + alt_2) \sin \lambda}{\sin \theta_1}$$

Elevation Angle.

Next, we can take advantage of the trigonometry of the scenario, we can leverage the Law of Sines to find the angles between each platform and the center of Earth. For either platform. Recall that λ is the inner-Earth angle between the platforms:

$$\sin \theta = \frac{(r_e + alt) \sin \lambda}{D}$$

which we can use to solve for θ .

Observe that in the platform-centric reference frames, θ and el are compliments of each other in a right-angle, as seen in Figure 7. Therefore, we can simply find el

$$el = \frac{\pi}{2} - \theta$$

Now, this does require a bit more logic. We need figure out the proper sign (+/-) of our angles. This indicates if each platform is looking above or below their respective local horizons' (ENU coordinate system) at the other platform.

From the diagram above, we can now see that if $r_e + alt < K$ then, $el = |el|$, and if $r_e + alt > K$, then $el = -|el|$. In the (very unlikely) case that $r_e + alt = K$ precisely, then it must be the case that $el = 0$.

Azimuth.

Once again taking advantage of the Spherical Law of Cosines, we can find

$$\begin{aligned} \cos \phi_1 &= \frac{\sin lat_2 - \cos \lambda \sin lat_1}{\sin \lambda \cos lat_1} \\ \cos \phi_2 &= \frac{\sin lat_1 - \cos \lambda \sin lat_2}{\sin \lambda \cos lat_2} \end{aligned}$$

Again, we need to account for the sign (+/-) of the azimuth. Note that on Earth, -180° and 180° are the same place, but approached from either east or west. So, if $\text{lon}_1 < \text{lon}_2$, then $\phi_1 = |\phi_1|$ and $\phi_2 = -|\phi_2|$. And, if $\text{lon}_1 > \text{lon}_2$, then $\phi_1 = -|\phi_1|$ and $\phi_2 = |\phi_2|$. In the (very unlikely) case that $\text{lon}_1 = \text{lon}_2$ precisely, the azimuths are 0° and 180° , respectively.

0.1 Az-El Calculation Method #2: Rotation from ECEF Coordinate Frame.

There is a more elegant way to calculate the azimuth and elevation angles from one platform to the other, going directly from ECEF to ENU, and simply taking into account the rotational orientation of each platform (w.r.t the other) in the form of matrices.

In ECEF, each platform can be represented by the vector $(x, y, z)^T$. Then, we can account for the rotation from one coordinate axis to the other using the shift (rotation) matrices. We will need two rotations to convert from ECEF to ENU coordinates. In order those are:

1. A clockwise rotation from Earth's east-axis by $90 - \text{lat}$ to align with the up-axis in ENU.
2. A clockwise rotation about the z-axis by $90 + \text{lon}$ to align the east-axis with the x-axis

The transformation between terrestrial frames can be defined as follow:

$$R_1[\alpha] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \alpha & \sin \alpha \\ 0 & -\sin \alpha & \cos \alpha \end{bmatrix}, R_2[\alpha] = \begin{bmatrix} \cos \alpha & 0 & -\sin \alpha \\ 0 & 1 & 0 \\ \sin \alpha & 0 & \cos \alpha \end{bmatrix}, R_3[\alpha] = \begin{bmatrix} \cos \alpha & \sin \alpha & 0 \\ -\sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

So, we can write the needed transformations in matrix notation as

$$R_1[-(\frac{\pi}{2} - \text{lat})]$$

$$R_3[-(\frac{\pi}{2} + \text{lon})]$$

and

$$R_3[-(\frac{\pi}{2} + \text{lon})]R_1[-(\frac{\pi}{2} - \text{lat})] = \begin{bmatrix} -\sin \text{lon} & -\cos \text{lon} \sin \text{lat} & \cos \text{lon} \sin \text{lat} \\ \cos \text{lon} & -\sin \text{lon} \sin \text{lat} & \sin \text{lon} \cos \text{lat} \\ 0 & \cos \text{lat} & \sin \text{lat} \end{bmatrix}$$

The unit vectors for east, north, and up in ENU coordinates, expressed using the platforms LLA is therefore

$$\hat{\mathbf{e}} = (-\sin \text{lon}, \cos \text{lon})$$

$$\hat{\mathbf{n}} = (-\cos \text{lon} \sin \text{lat}, -\sin \text{lon} \sin \text{lat}, \cos \text{lat})$$

$$\hat{\mathbf{u}} = (\cos \text{lon} \cos \text{lat}, \sin \text{lon} \cos \text{lat}, \sin \text{lat})$$

In order to convert the ENU vectors to azimuth and elevation, we simply take the LOS vector, given by the ECEF coordinates of each platform [4] [1] [2]. That is

$$\hat{\mathbf{L}} = \frac{\begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} - \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix}}{\left\| \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} - \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix} \right\|}$$

Now, we can see that the azimuth and elevation in the local ENU system are given by

$$\hat{\mathbf{L}} \cdot \hat{\mathbf{e}} = \cos(el) \sin(az)$$

$$\hat{\mathbf{L}} \cdot \hat{\mathbf{n}} = \cos(el) \cos(az)$$

$$\hat{\mathbf{L}} \cdot \hat{\mathbf{u}} = \sin(el)$$

or

$$el = \sin^{-1}(\hat{\mathbf{L}} \cdot \hat{\mathbf{u}})$$

$$az = \tan^{-1}\left(\frac{\hat{\mathbf{L}} \cdot \hat{\mathbf{e}}}{\hat{\mathbf{L}} \cdot \hat{\mathbf{n}}}\right)$$

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